CSE 471: MACHINE LEARNING

Learning Probabilistic Models

Bayesian Learning

h_1: 100% cherry,
h_2: 75% cherry + 25% lime,
h_3: 50% cherry + 50% lime,
h_4: 25% cherry + 75% lime,
h_5: 100% lime.

$$P(h_i|\mathbf{d}) = \alpha \boxed{P(\mathbf{d}|h_i)}\boxed{P(h_i)}$$

Hypothesis prior

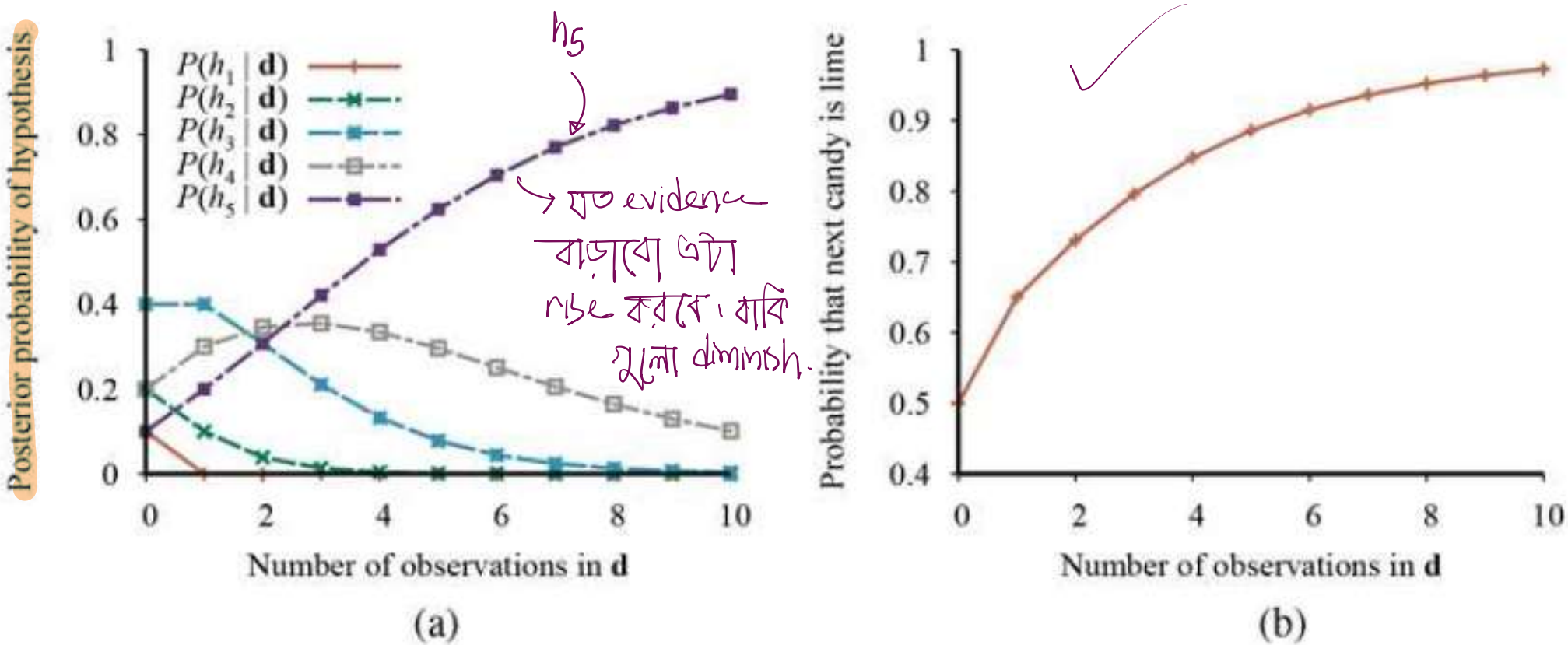
Likelihood

$$P\left(\mathbf{d}\middle|h_i\right) = \prod_j P\left(d_j\middle|h_i\right)$$

$$\boxed{\mathbf{P}\left(X\middle|\mathbf{d}\right)} = \sum_i \mathbf{P}\left(X\middle|h_i\right) P\left(h_i\middle|\mathbf{d}\right)$$

Prediction

Bayesian Learning



(a) Posterior probabilities $P(h_i|d_1, \ldots, d_N)$ from Equation (20.1). The number of observations N ranges from 1 to 10, and each observation is of a lime candy. (b) Bayesian prediction $P(D_{N+1} = lime|d_1, \ldots, d_N)$ from Equation (20.2).

Maximum a posteriori hypothesis

- Single most probable hypothesis
- An h_i that maximizes $P(h_i | d)$
- Optimization problem, rather than summation
  - Often times, it is much easier to solve

Maximum-Likelihood hypothesis

- Uniform prior assumed over the hypothesis space
- Find the h_i that maximizes $P(d \mid h_i)$

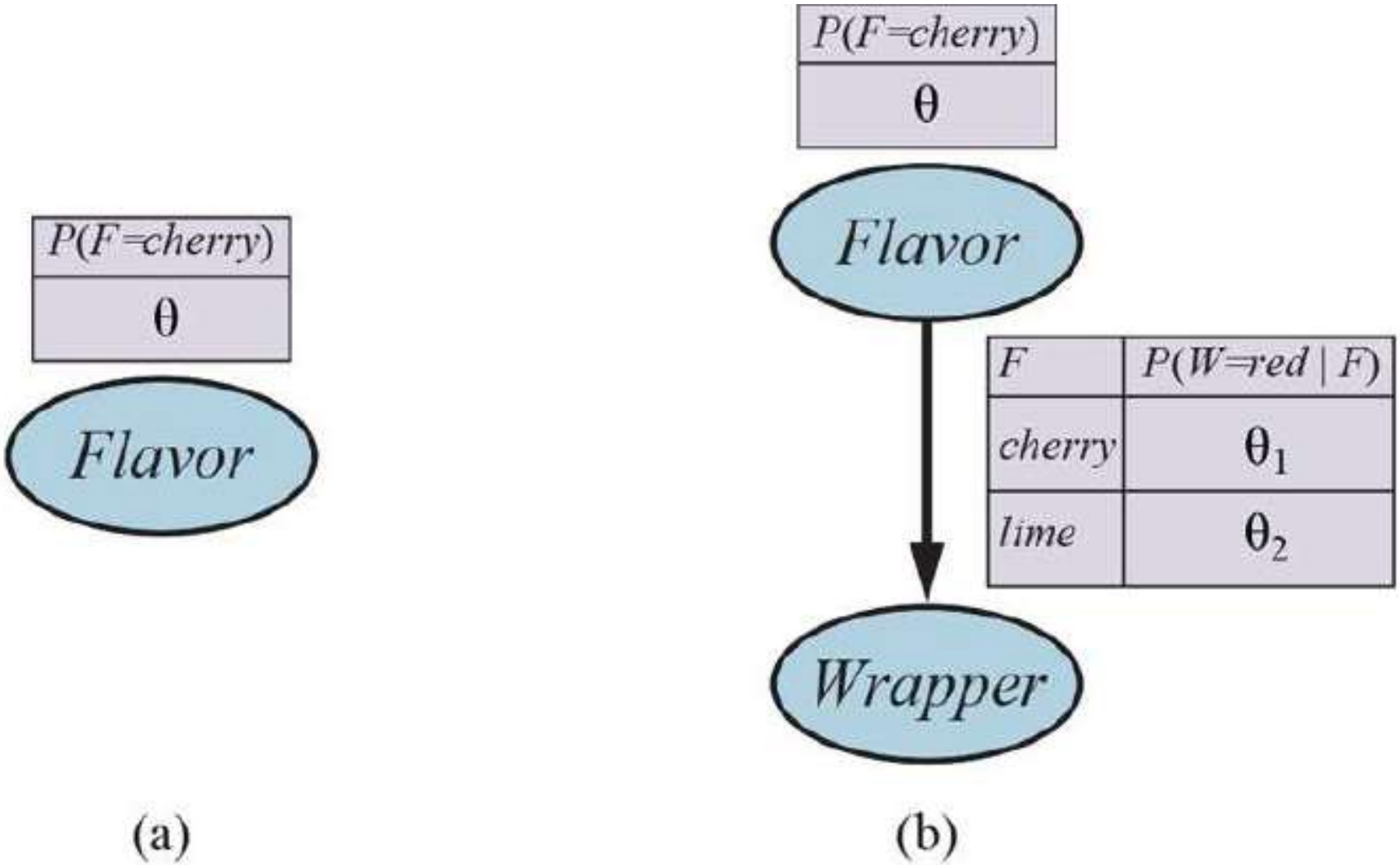
ML parameter learning

$$P\left(\mathbf{d}\middle|h_0\right) = \prod_{j=1}^{N} P\left(d_j\middle|h_0\right) = \theta^c \cdot \left(1-\theta\right)^{\ell}.$$

$$L\left(\mathbf{d}\middle|h_0\right) = \log P\left(\mathbf{d}\middle|h_0\right) = \sum_{j=1}^{N} \log P\left(d_j\middle|h_0\right) = c\log\theta + \ell\log\left(1-\theta\right)$$

$$\frac{dL(\mathbf{d}|h_0)}{d\theta} = \frac{c}{\theta} - \frac{\ell}{1-\theta} = 0 \qquad \Rightarrow \qquad \theta = \frac{c}{c+\ell} = \frac{c}{N}.$$

ML parameter learning



(a) Bayesian network model for the case of candies with an unknown proportion of cherry and lime. (b) Model for the case where the wrapper color depends (probabilistically) on the candy flavor.

ML parameter learning

$$P\left(\mathbf{d}\middle|h_{\theta,\theta_1,\theta_2}\right) = \theta^c\left(1-\theta\right)^{\ell}\cdot\theta_1^{r_c}\left(1-\theta_1\right)^{g_c}\cdot\theta_2^{r_\ell}\left(1-\theta_2\right)^{g_\ell}$$

$$L = [c\log\theta + \ell\log(1-\theta)] + [r_c\log\theta_1 + g_c\log(1-\theta_1)] + [r_\ell\log\theta_2 + g_\ell\log(1-\theta_2)]$$

$$\begin{aligned}
\frac{\partial L}{\partial\theta} &= \frac{c}{\theta} - \frac{\ell}{1-\theta} = 0 &\Rightarrow\quad \theta = \frac{c}{c+\ell}\\
\frac{\partial L}{\partial\theta_1} &= \frac{r_c}{\theta_1} - \frac{g_c}{1-\theta_1} = 0 &\Rightarrow\quad \theta_1 = \frac{r_c}{r_c+g_c}\\
\frac{\partial L}{\partial\theta_2} &= \frac{r_\ell}{\theta_2} - \frac{g_\ell}{1-\theta_2} = 0 &\Rightarrow\quad \theta_2 = \frac{r_\ell}{r_\ell+g_\ell}
\end{aligned}$$

ML parameter learning (Continuous model)

$$P\left(x\right) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{\left(x-\mu\right)^2}{2\sigma^2}}$$

$$L = \sum_{j=1}^{N} \log \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{\left(x_j-\mu\right)^2}{2\sigma^2}} = N\left(-\log\sqrt{2\pi} - \log\sigma\right) - \sum_{j=1}^{N} \frac{\left(x_j-\mu\right)^2}{2\sigma^2}$$

$$\frac{\partial L}{\partial \mu} = -\frac{1}{\sigma^2}\sum_{j=1}^{N}\left(x_j-\mu\right) = 0 \qquad \Rightarrow \qquad \mu = \frac{\sum_j x_j}{N}$$

$$\frac{\partial L}{\partial \sigma} = -\frac{N}{\sigma} + \frac{1}{\sigma^3}\sum_{j=1}^{N}\left(x_j-\mu\right)^2 = 0 \qquad \Rightarrow \qquad \sigma = \sqrt{\frac{\sum_j\left(x_j-\mu\right)^2}{N}}$$

Conditional distribution P(Y | X)

- Y has a Gaussian distribution
  - Fixed standard deviation
  - Mean is linearly dependent on X

$$P\left(y \middle| x\right) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{\left(y-\left(\theta_1 x+\theta_2\right)\right)^2}{2\sigma^2}}$$

- Maximizing the log likelihood with respect to these parameters is the same as *minimizing:*

$$\left(y - \left(\theta_1 x + \theta_2\right)\right)^2$$